INTERNATIONAL SCHOOL FOR YOUNG ASTRONOMERS

UNIVERSIDAD YACHAY TECH

PYTHON FOR ASTROPHYSICS

Lecture 6

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Lecture 6 goals:

1. Learn the capabilities of pyvista.

2. Create animations of a full simulation dataset (an MHD vortex).

3. Visualise simulation files with matplotlib and plotly.

What do you need for the practicals?

- A PC/laptop with any OS.

- Internet access.

- A Google/gmail account.

- A GitHub account (desirable, not strictly needed).

The pyvista library

PyVista provides an intuitive Python interface for plotting and mesh analysis, built on top of VTK (Visualization Toolkit).

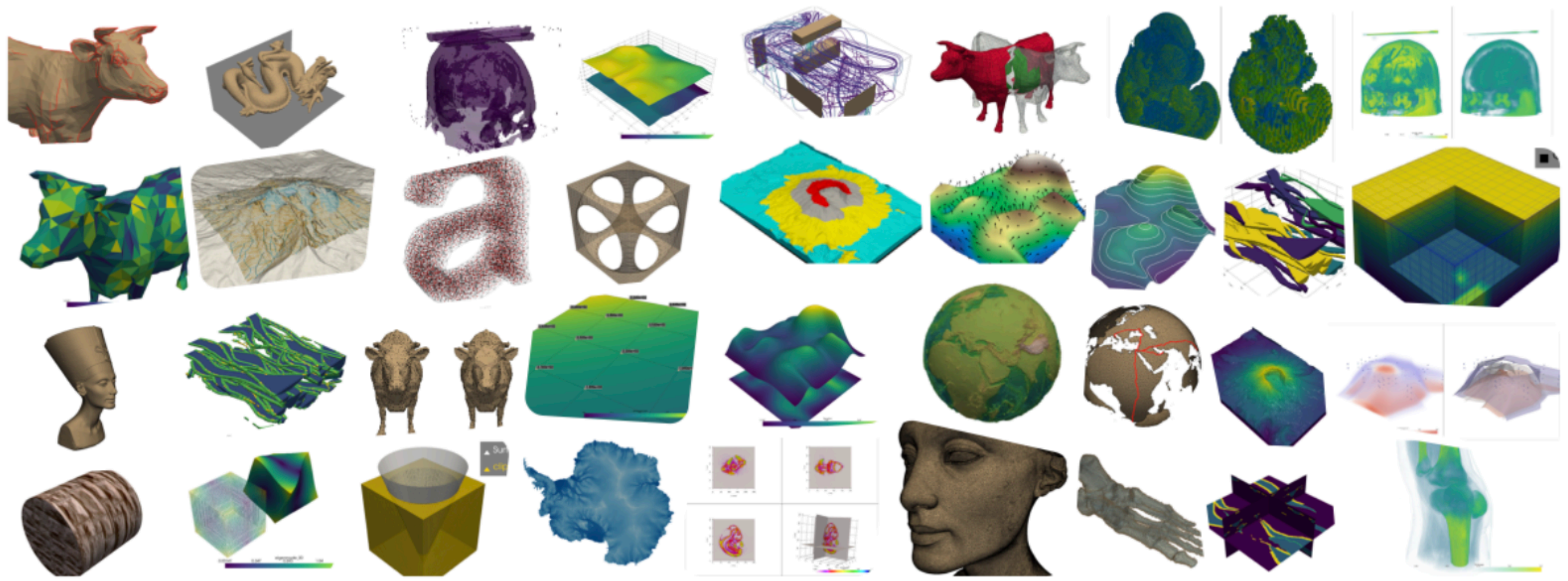
It supports several meshes including structured/unstructured grids, point clouds, and surface meshes with easy manipulation and filtering.

It integrates well with other scientific libraries like **NumPy** and **Matplotlib**.

We will use it for I/O of **VTK files.**

VTK = Visualisation Toolkit

3D plotting and mesh analysis through a streamlined interface for the Visualization Toolkit (VTK)



https://docs.pyvista.org/index.html

VTK data format

VTK: Visualisation ToolKit (VTK) format

This format is an open-source data format, developed by Kitware, and widely used in computational fluid dynamics and computer graphics applications.

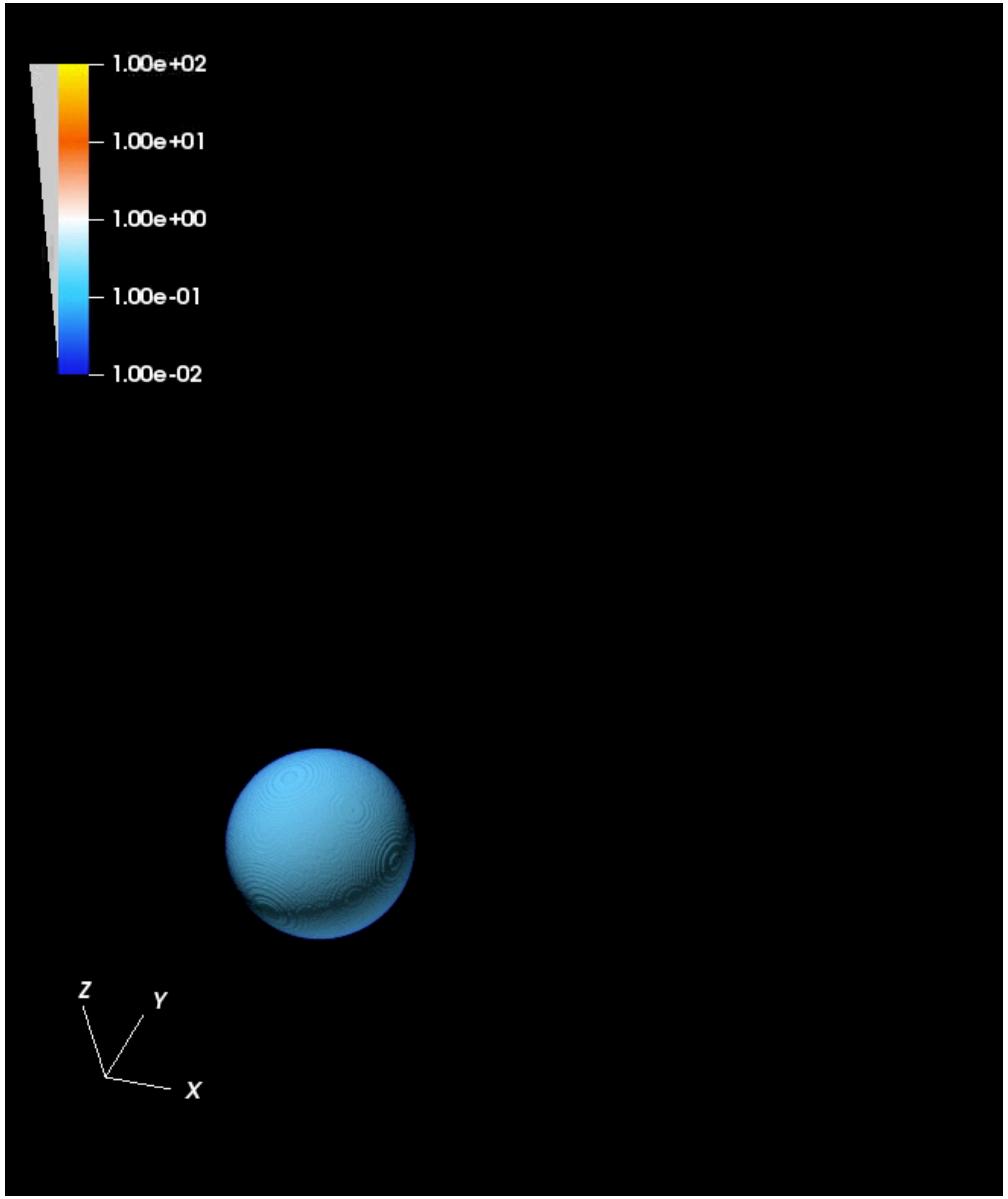
VTK file structure:

- File version and ID.
- Header
- Data type, which can be Binary or ASCII.

STRUCTURED_GRID

UNSTRUCTURED_GRID

RECTILINEAR_GRID



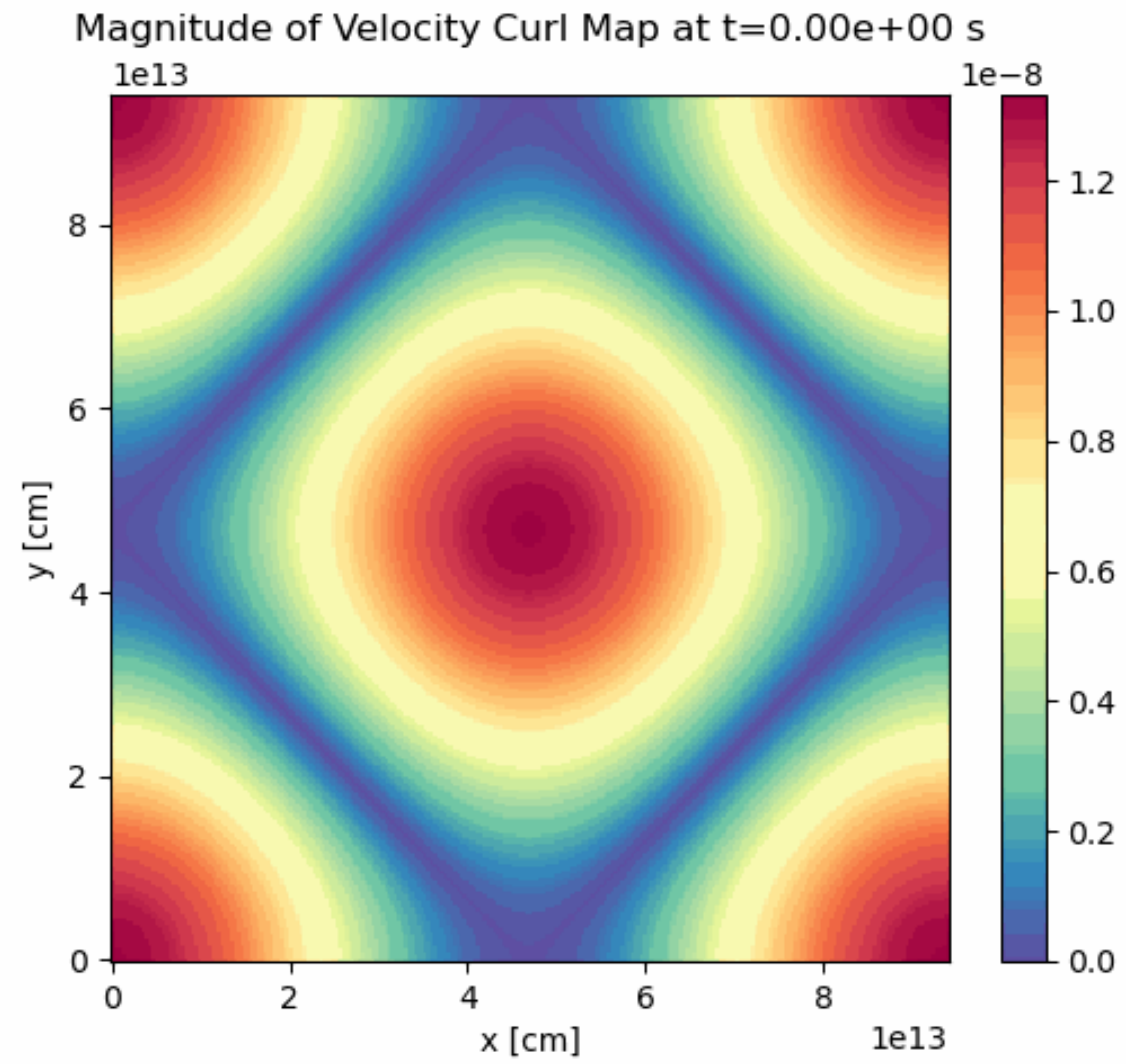
https://ui.adsabs.harvard.edu/abs/2016MNRAS.455.1309B

Orszag-Tang Vortex

A **doubly periodic fluid configuration** leading to 2D supersonic **magnetohydrodynamical** (MHD) turbulence

Benchmark for the comparison of MHD numerical solvers.

The computational domain is periodic box with dimensions: $[0,2\pi]^D$ where D=2 is the number of spatial dimensions.



In code units, the initial conditions are given by:

$$\vec{v} = \left(-\sin y,\ \sin x\right)\ , \qquad \overrightarrow{B} = \left(-\sin y,\ \sin 2x\right)\ , \qquad \rho = 25/9\,, \qquad p = 5/3$$

Tutorial Time

1. Please log into your gmail accounts:

2. Open this lecture on GitHub:

   https://github.com/wbandabarragan/ISYA2025/blob/main/Python_for_Astrophysics/4_tutorial_py4astro.ipynb

3. Click on the “**Open in Colab**” icon and you are ready to code!